American Computer Science League

2020-2021 ● Contest 1: Shorts ● Junior Division

1. Computer Number Systems

How many 1's are there in the binary representation of the decimal numbers from 22 to 32, inclusive?

A. 34
B. 35
C. 36
D. 37
E. 38

2. Computer Number Systems

Which number is the largest?

101100000_2 535_8 $15F_{16}$

A. 101100000_2
B. 535_8
C. $15F_{16}$
D. 2 of them are the largest
E. all 3 are the same

3. Recursive Functions

Find $f(7)$, given:

$$f(x) = \begin{cases} 1 & if\ x = 1 \\ x + f(x-1) & if\ x > 1 \end{cases}$$

A. 27
B. 28
C. 30
D. 36
E. 38

4. Recursive Functions

Find $f(20)$, given:

$$f(x) = \begin{cases} f\left(\frac{x}{2}\right) + 1 & \text{if } x \text{ is even} \\ f(x+3) - 1 & \text{if } x \text{ is odd and } x > 3 \\ \frac{x+1}{2} & \text{otherwise} \end{cases}$$

A. 2
B. 3
C. 4
D. 5
E. 6

5. What Does This Program Do? (Branching)

What is the output when this program is executed?

```
a = 24 : b = 2 : c = 3
if a / c  ==  b ^ c then
        b = b * c
end if
if c + a / b > b then
        a = a - b * c
end if
b  =  b + a / 3
if  b  > a +  c  then
        b = 2 * b
else
        c = a / c
end if
output a + b / c ^ 2.
```

A. 8
B. 10
C. 22
D. 28
E. 49

American Computer Science League

2020-2021 ● Contest 1: Shorts ● Intermediate Division

1. Computer Number Systems

How many 1's are there in the binary representation of 2021_{10} ?

A. 6
B. 7
C. 8
D. 9
E. 10

2. Computer Number Systems

Evaluate and express the answer in hexadecimal:

$72126_8 - 54127_8$

A. $1BFF_{16}$
B. $1AEF_{16}$
C. $1BFE_{16}$
D. $1AFF_{16}$
E. BFF_{16}

3. Recursive Functions

Find $f(10)$, given:

$$f(x) = \begin{cases} 2x+2 & if\ x \le 2 \\ f(x-1) - f(x-2) & if\ x > 2 \end{cases}$$

A. -6
B. -4
C. 2
D. 4
E. 6

4. Recursive Functions

Find $f(f(17))$, given:

$$f(x) = \begin{cases} 2+f(x-2) & if\ x \ge 8 \\ x-4 & if\ x < 8 \end{cases}$$

A. 9
B. 11
C. 13
D. 15
E. 1

5. What Does This Program Do? (Branching)

How many different unique values do the variables a, b, c, and d take on when this program is executed?

```
a = 5 : b = 2 : c = 25 : d = 4
if c / a > d then
        b = b + d
else
        b = b + c / a
end if
c = c − a * b / 2
if (b < c) && (d < a) then
        d = d / 4
else
        a = c ^ 2 / a
end if
if d % a == 0 then
        d = b + c
end if
if (c /2 > b) || ( b − d == a) then
        d = 5 * d
end if
```

A. 4
B. 5
C. 6
D. 7
E. 8